Time Series Prediction of Stock Returns Using Machine Learning Models: Study on Dow Jones Industrial Average Stocks (2000 – 2024)

An extended research project report submitted to the University of Manchester for the degree of

Master of Science in Data Science

in the Faculty of Humanities

Year of submission

2025

Student ID

10581125

School of Social Sciences

TABLE OF CONTENTS

ABSTRACT **4**
DECLARATION **5**
INTELLECTUAL PROPERTY STATEMENT **6**
ACKNOWLEDGEMENTS **7**
1. INTRODUCTION **8**
2. LITERATURE REVIEW **11**
2.1 Traditional Models and Machine Learning Approaches 11
2.2 Machine Learning Models and Predictive Performance 12
2.3 Use of Technical Indicators in Stock Return Prediction 13
2.4 Comparison of ML Models for Stock Return Prediction 15
3. METHODOLOGY **17**
3.1 Data Collection and Resources 17
3.2 Target Prediction: Monthly Return 17
3.3 Feature Engineering and Technical Indicators 18
3.4 Data Preprocessing 19
3.5 Sectoral Classification 20
3.6 Modeling Techniques 21
3.7 Model Evaluation 23
4. RESULTS **25**
4.1 Overall Performance Comparison 25
4.2 Role of Technical Indicators 26
4.3 Stock-Level Conditions and Model Performance 27
4.4 Stock and Metric Comparisons 33
4.5 Sector-Level Results 36
5. DISCUSSION **38**
6. CONCLUSION AND FUTURE WORK **40**
REFERENCES **41**
APPENDIX **43**
A. Stock-Based Model Results 43

B. Sector-Based Model Results 58
C. ARIMA 60

LIST OF TABLES

Table 1: DJIA Companies 17
Table 2: DJIA Sectors and Companies 21
Table 3: Overall Model Performances 25
Table 4: Mean and Std. of Monthly Return for Selected Stocks for Classification 27
Table 5: PG Model Results 28
Table 6: JNJ Model Results 29
Table 7: NVDA Model Results 30
Table 8: BA Model Results 30
Table 9: CAT Model Results 31
Table 10: MSFT Model Results 32
Table 11: Overall Sector Performances 36

LIST OF FIGURES

Figure 1: VZ LSTM Actual vs. Predicted Graph 33
Figure 2: JNJ RF (All Features) Actual vs. Predicted Graph 34
Figure 3: BA SVR (All Features) Actual vs. Predicted Graph 35

Total word count: 7679

ABSTRACT

This extended research project examines the predictability of companies included in the Dow Jones Industrial Average (DJIA) index using machine learning techniques using data from 2000 to 2024. The study derived monthly returns from daily price and volume data only, and technical indicators such as moving averages, momentum, RSI, and volatility were used to support and test the machine learning models. Random Forest, XGBoost, Support Vector Regression, and Long-Short Term Memory models were used for machine learning. Model performance was evaluated by using root mean square error (RMSE) and mean absolute error (MAE), and in addition to these metrics, the directional accuracy metric was used to examine the directional forecasting performance of the models. The study also compared the predictability of sectors within the DJIA stocks. The study's findings indicate that machine learning methods exhibit varying levels of predictive success, demonstrating varying performance across different sectors and stocks. Tree-based Random Forest and XGBoost models generally performed better in minimizing RMSE and MAE metrics, while the LSTM model produced stronger directional prediction results in more stable and low-volatility stocks. In high-volatility stocks, the Support Vector Regression model improved model performance by benefiting more from lagged returns. Furthermore, results demonstrate that no single model can be considered absolutely accurate, and that different models can be successful in predicting monthly returns for stocks with different characteristics. By assessing the contribution of technical indicators derived from price and volume data to predictive power, the study demonstrates the potential of machine learning methods for investment applications.

DECLARATION

No portion of the work referred to in this extended research project report has been submitted in support of an application for another degree or qualification of this or any other university or other institute of learning.

INTELLECTUAL PROPERTY STATEMENT

i. The author of this extended research project report (including any appendices and/or schedules to this report) owns certain copyright or related rights in it (the "Copyright") and s/he has given The University of Manchester certain rights to use such Copyright, including for administrative purposes.

ii. Copies of this report, either in full or in extracts and whether in hard or electronic copy, may be made only in accordance with the Copyright, Designs and Patents Act 1988 (as amended) and regulations issued under it or, where appropriate, in accordance with licensing agreements which the University has entered into. This page must form part of any such copies made.

iii. The ownership of certain Copyright, patents, designs, trademarks, and other intellectual property (the "Intellectual Property") and any reproductions of copyright works in the report, for example graphs and tables ("Reproductions"), which may be described in this report, may not be owned by the author and may be owned by third parties. Such Intellectual Property and Reproductions cannot and must not be made available for use without the prior written permission of the owner(s) of the relevant Intellectual Property and/or Reproductions.

iv. Further information on the conditions under which disclosure, publication and commercialisation of this report, the Copyright and any Intellectual Property and/or Reproductions described in it may take place is available in the University IP Policy (see https://documents.manchester.ac.uk/display.aspx?DocID=24420), in any relevant dissertation restriction declarations deposited in the University Library, The University Library's regulations (see https://www.library.manchester.ac.uk/about/regulations/) and in The University's Guidance for the Presentation of dissertations.

ACKNOWLEDGEMENTS

I would like to express my sincere respect and gratitude to my extended research project supervisor, Dr. Lijie Yu, who supported me at every stage of the preparation of this study with her sincerity, patience and valuable knowledge.

I am also grateful to my sister and to my family, whose unconditional encouragement and support have been with me at every stage of my studies.

1. INTRODUCTION

Successfully and accurately predicting stock returns has been a long topic of discussion and aspiration in both academic finance literature and applied investment management. Even a small improvement in the estimation of stock returns can help investors and portfolio managers become more efficient in hedging against risks and determining strategies. As capital markets become increasingly volatile, complex, and sophisticated, predicting future price movements, particularly through automated and data-driven methods, is considered a significant competitive advantage.

The fundamental challenge of predicting stock returns stems from the stochastic and dynamic nature of financial markets. Prices are influenced by many different factors, including macroeconomic indicators, company fundamentals and characteristics, investor characteristics, and external economic fluctuations. These factors produce data structures that are difficult to track and highly complex, with noise. Financial time-series data structures challenge both statistical and algorithmic forecasting methods. Many studies on stock return forecasting, particularly The Efficient Markets Hypothesis (EMH), argue that historical information is already reflected in prices, making systematic and consistent forecasting impossible (Fama, 1970). However, recent research, along with the development of artificial intelligence and machine learning (ML), has suggested that models have emerged that challenge this hypothesis and improve prediction accuracy when used with various indicators and methods.

There has been evidence that technical indicators derived from historical price data can increase the predictive power of models. Technical indicators such as Relative Strength Index (RSI), which measure the speed and magnitude of current price changes to identify overbought or oversold conditions, momentum, and moving average provide meaningful insights into ML methods that characterize stock behavior and, in line with these characteristics, whether the stock price is trending upwards, downwards, or remaining stable. These indicators are widely used in practice and studied in academic studies, believed to reflect investor behavior and market sentiment (Gencay and Stengos, 1998), and are considered data-driven indicators representing price momentum and trends.

The rise of predictive and computational power in parallel with financial theory has led researchers to alternative modeling. With the development of ML algorithms such as Random Forest, Support Vector Regressor (SVR), and XGBoost are increasingly being adopted for their ability to learn the complexity and nonlinear relationships in multidimensional data (Ballings, 2015). These models, unlike traditional methods like ARIMA, differ from traditional time-series models in that they do not require assumptions such as linear structure or stationarity in financial datasets, offering models with high predictive potential in research.

Likewise, LSTM (Long Term Short Memory) which is a deep learning (DL) model also offers powerful capabilities for capturing features such as time dependency and seasonality in time-series data. LSTMs have demonstrated successful results in financial forecasting tasks thanks to their ability to capture long-term dependencies (Hansson, 2017).

Despite the increasing interest in these models and the popularity of stock return forecasting studies, studies systematically comparing traditional statistical methods with modern ML approaches and demonstrating their predictive performance are still limited, particularly those based solely on time-series data and technical indicators for stock-based forecasts. This study aims to fill this gap by forecasting stock returns using various ML methods using only time-series data and supplementing it with technical indicators derived from this data.

This thesis focuses on forecasting the monthly returns of 30 companies within the Dow Jones Industrial Average (DJIA) between 2000 and 2024. This study is particularly limited in scope to time-series data and utilizes technical indicators derived from this data.

The aim of this study is to examine and compare the forecast performance of ML models. Various technical indicators will also be used to evaluate the impact of indicators on performance and examine forecast accuracy differences across sectors. In this context, the traditional ARIMA model is mentioned in Appendix C, but study is limited to ML models. The study aims to answer the following questions:

1. Are machine learning techniques successful in predicting stock returns using historical time series data, which ML models have better performance?
2. How do technical indicators affect the predictability of stock returns?

3. Under what stock-level conditions do technical indicators offer stronger predictive power for returns?
4. Which DJIA sectors offer higher forecast performance?

In conclusion, this research aims to contribute to data-driven financial modeling studies, reveal which modeling techniques provide the highest success in stock return forecasting for DJIA stocks, and demonstrate the success of ML in financial forecasting with the importance of technical indicators. The study findings demonstrate that ML algorithms offer some predictive power, albeit not as strong, for DJIA stocks. While the RF and XGBoost models performed better in error metrics, LSTM provided higher directional accuracy in low-volatility stocks. In the sectoral analysis, the technology sector stands out for its directional prediction rate, while the healthcare sector stands out for its low error metrics. The main observation of the study is that different models perform better under different conditions.

2. LITERATURE REVIEW

Forecasting stock returns has been a controversial and challenging research topic in finance for many years. The topic is recognized as one of the most relevant but challenging tasks (Chen and Hao, 2017). The Efficient Market Hypothesis (EMH) denies that asset price movements can be predicted using historical information and posits their unpredictability (Fama, 1970). However, studies conducted with some anomalies related to past returns and developing ML methods have challenged EMH.

2.1 Traditional Models and Machine Learning Approaches

For years, traditional time series models, such as ARIMA, have been used to predict stock returns using variables such as volume and price from previous years. These models, while based on statistical theories, offer reasonable and interpretable results, but they also come with significant limitations. In particular, classical models like ARIMA typically assume linear relationships within the data and expect data to be stationary, making it difficult to capture the highly nonlinear and complex patterns often present in financial time series (Sun and Deng, 2025). These traditional methods may have difficulty capturing seasonality, complexity and interacting factors.

In recent years, ML techniques have become a powerful tool in financial forecasting with increasing computational power, helping ML to extract meaning from complex structures compared to traditional methods. (Gu et al., 2020). ML methods can capture nonlinear interactions by making flexible assumptions with capability of handling high-dimensional, complex relationships. The most used and researched ML methods in this field, such as Random Forest (RF), Gradient Boosted Trees (XGBoost), Support Vector Machines (SVM), and DL methods such as LSTM, can learn non-linear, complex relationships and interaction effects from data (Kumbure et al., 2022). These models prevent overfitting through techniques such as cross-validation and reveal trends that traditional regression cannot capture (Leippold et al., 2022). While traditional models rely on predefined economic assumptions, ML methods use a data- and pattern-driven approach to learn hidden relationships in price movements and provide powerful predictions (Fischer and Krauss, 2018).

2.2 Machine Learning Models and Predictive Performance

One of the central research points in the literature is whether ML techniques can provide better forecast accuracy or profitability than statistical and conventional approaches for predicting stock returns. A growing body of research demonstrates that ML-based models outperform traditional models under many circumstances. Gu et al. (2020) conducted a study comparing a wide range of ML techniques for stock returns. They report that these ML forecasts achieve significantly higher out-of-sample predictive power compared to previous linear models, translating into substantial economic gains for investors. In this study, a neural network portfolio strategy significantly improved the performance of strategies based on ML predictions, achieving a significantly higher Sharpe ratio compared to a simple buy-sell strategy. The results demonstrate that ML models can capture signals missed by traditional methods, leading to superior investment performance.

Other studies support these findings. Choi et al. (2023) applied market-specific ML models to predict stock returns in international markets and found that the ML models significantly outperformed linear models in out-of-sample tests. Their results showed that ML methods capturing nonlinear patterns resulted better predictions and higher portfolio returns than ordinary least squares regression in almost every market. They suggested that deep neural networks and tree-based models tended to exhibit high performance.

Applications for ML in directional forecasting and trading have also showed potential. Hansson (2017) forecasted the direction of stock index returns using LSTM neural networks and contrasted them with a traditional ARMA-GARCH model. He discovered that the ML method was noticeably better at forecasting the direction of price movements (up or down), even if the LSTM's numerical return forecasts were comparable to those of the statistical model. In the case of a smaller, less efficient market (Sweden’s OMX), the LSTM achieved significantly accurate direction forecasts. Even for larger markets where directional predictability was weaker, deploying a trading strategy based on the LSTM’s signals vastly outperformed a strategy based on the ARMA-GARCH model. These results highlight that even if ML and traditional models generate similar errors in a regression sense, ML’s ability to capture non-linear dynamics can translate into better market timing and higher trading returns.

However, it is important to balance the optimism of ML methods with factors such as market conditions and data structure dependence. Some research suggests that ML models perform well in niche market segments or during periods of low market efficiency. It has been suggested that ML methods often profit from small companies that are difficult to arbitrage (Avramov et al., 2023). The predictive performance of ML models deteriorates significantly when more realistic economic constraints are applied. Furthermore, the absolute predictive accuracy of ML models is also limited. For example, out-of-sample R^2 values of return forecasts are only a few percentage points, indicating that a significant portion of returns is still random. Nevertheless, even a small directional improvement can be economically significant. In conclusion, ML is a useful tool for predicting stock returns, but these models can be context dependent.

International level research also highlights the importance of markets. ML models have demonstrated significant predictive performance in the Chinese stock market, known for its concentration of retail investors and diverse regulatory frameworks. Leippold's (2022) study built a comprehensive ML-based model for the Chinese market and found that this model achieved economically meaningful performance even after accounting for transaction costs. Interestingly, the predictability of the Chinese market exhibited some unique patterns. Short-term return forecasts were particularly strong for small-cap stocks with a large retail investor base, and even large state-owned enterprises demonstrated predictability in long-term forecasts. These studies demonstrate that ML algorithms can detect complex and behavioral patterns across diverse market structures and behaviors. Generally, this complexity demonstrates that ML methods outperform traditional methods in behavioral and seasonal data, making them a useful tool for stock return forecasting. However, this situation varies across different markets and market conditions.

2.3 Use of Technical Indicators in Stock Return Prediction

One of the key advantages of ML in finance is the integration of various technical indicators derived from historical prices into models and their use as independent variables for forecasting. While traditional models are limited to a few variables due to collinearity limitations and overfitting issues with small data, ML models can handle many variables and identify which ones are important for target prediction. Technical indicators used in stock market forecasting

studies are among the most popular input features (Kumbure et al., 2022). Among these, moving averages (MA20, MA50, which smooth out price fluctuations and capture medium- to long-term trends), momentum (the difference between current and past prices), the Relative Strength Index (RSI, a momentum oscillator that measures the speed and magnitude of price changes), volatility (a statistical measure of dispersion often standard deviation of returns) are often used as input variables in ML algorithms because they represent formations in the markets.

The importance of these technical indicators has also been emphasized in various studies. When ML algorithms were used to forecast US stocks, momentum-related variables were found to be the strongest signals (Gu et al., 2020). Analyses suggest that factors related to momentum, liquidity, and volatility provide stability in predicting future returns. However, as previously mentioned, the importance of different technical indicators can vary in different contexts.

From this perspective, in Chinese stock market, liquidity measures such as trading volume ratios and illiquidity indices have been shown to be the most effective predictors (Leippold et al., 2022). This divergence of the Chinese market from the US stock market can be explained by the high prevalence of small investors and the increased predictive power of periods of low liquidity. Considering these studies, an indicator that works well in one market may not be as important as another.

In addition to being used as input for model estimation, technical indicators can also help understand why a model is successful. For example, one can analyze under which market conditions signals such as volatility or moving averages increase forecast accuracy. It has been suggested that some technical signals are more useful in certain market regimes or sectors (Rapach et al., 2018).

In summary, technical indicators such as momentum, moving averages, RSI, and volatility can significantly contribute to forecasting power because they convey information about market sentiment, stock and price trends. ML models built with technical indicators can generate multidimensional and powerful forecasting structures by evaluating classic technical analysis signals with modern data science methods (Kumbure et al., 2022).

2.4 Comparison of ML Models for Stock Return Prediction

With the development of ML methods and their use in stock return prediction, questions such as "Which models perform better?" have arisen. While the answer to this question depends on the data structure and market characteristics, general trends have also been observed based on the literature review. Simpler ML methods include versions combined with linear regression regularization and methods such as kernel-based SVM, while more complex models feature nonlinear structures such as ensemble decision trees and deep neural networks (Kumbure et al., 2022).

Basic neural networks and algorithms like SVR have been preferred due to their ability to capture structures not inherent in financial data. With increasing processing power and complexity over time, multilayer deep networks and time-series-focused recurrent networks have become more common (Althelaya et al., 2018). Furthermore, the use of hybrid models and ensemble models is increasingly preferred such as identifying significant variables with the XGBoost model and modeling time dependencies with a neural network are examples of hybrid models (Kim and Won, 2018).

When algorithms are directly compared, many studies show that models that can capture nonlinear and complex patterns are more successful. According to Gu's study, extensive experiments involving linear models, tree-based models, and neural networks suggest that the best-performing models are tree-based structures and deep neural network models (Gu et al., 2020). Surprisingly, while all models identified similar variables, only nonlinear models effectively learned the interactions of these signals and provided more consistent predictions.

Another international study by Choi et al. (2023) also supports the general findings. In the study, both neural networks and regression trees produced higher R^2 and portfolio returns than linear models in most contexts. However, as previously mentioned, it is important to remember that complex models like DL networks carry the risk of overfitting in markets with limited observations, and that models can vary depending on the market (Choi et al., 2023).

Recurrent neural networks such as LSTM are popularly used in stock prediction due to their ability to model time dependencies and to prioritize information. These models have been observed to outperform both traditional and some ML models (Fischer and Krauss, 2018).

Furthermore, LSTM networks have been found to have higher predictive performance in market timing compared to the ARMA-GARCH model (Hansson, 2017). However, for deep neural networks to be successful, hyperparameter tuning must be carefully adjusted; parameter selection and the number of hidden layers significantly affect prediction accuracy (Yadav et al., 2020).

Ensemble methods such as Random Forest and XGBoost have been found to be quite successful in predictions. These tree-based models have the capacity to naturally distinguish which indicators are important and trends such as seasonality (Freyberger et al., 2020).

In summary, the consensus in the literature indicates that algorithms that capture non-linear interactions offer higher predictive power than simple, traditional methods. While DL models provide high accuracy, ensemble methods are powerful alternatives for different market conditions. Complex models are more advantageous in dynamic markets, while simple models are more advantageous in efficient markets; thus, no model is superior in all circumstances. Therefore, while ML has strong potential for return prediction, its success depends on market conditions.

The literature demonstrates the potential of ML models in stock return forecasting. However, existing studies largely focus on daily data, macroeconomic or sentiment-based variables, lack sector comparisons, and do not directly examine DJIA stocks. This study aims to fill this gap by evaluating the importance of technical indicators, comparing model performance with sector-based comparisons, and focusing on monthly returns on DJIA stocks.

3. METHODOLOGY

This study covers 30 DJIA companies from January 2000 to December 2024. The DJIA is one of the most recognized indicators of the US stock market, and includes large, established, and international companies from a variety of sectors. Therefore, it provides an appropriate sample in terms of market representativeness.

3.1 Data Collection and Resources

Using only historical price data, daily open, close, high, low, and volume were collected using the Python library "yfinance," and each company's data was saved in separate CSV files. A single Python script automated the extraction. Companies listed in DJIA index can be found below (Table 1):

Table 1: DJIA Companies

3M	Boeing	Disney	Johnson & Johnson	Nike	Travelers
American Express	Caterpillar	Goldman Sachs	JPMorgan Chase	Nvidia	UnitedHealth Group
Amgen	Chevron	Home Depot	McDonald's	Procter & Gamble	Verizon
Amazon	Cisco	Honeywell	Merck	Salesforce	Visa
Apple	Coca-Cola	IBM	Microsoft	Sherwin-Williams	Walmart

3.2 Target Prediction: Monthly Return

The main variable to be estimated is the monthly stock return of each company (r_t+1) and is calculated according to the formula below.

$$r_{\{t+1\}} = \frac{\{P_{\{close,t+1\}} - P_{\{close,t\}}\}}{\{P_{\{close,t\}}\}}$$

This transformation played a role in converting the financial time series into a target variable that will be estimated in the form of monthly percentage returns based on the closing prices on the last day of each month.

3.3 Feature Engineering and Technical Indicators

In this study, technical indicators and lagged variables derived from price data were created to analyze the predictive power of models and price behavior. The selection of these technical indicators was based on popular indicators widely used in literature and financial analysis, which provide information on price momentum, trend direction, and volatility.

Technical Indicators

- **Moving Average-20 (MA20):** 20-day average of daily closes, was calculated for the last day of each month to capture short trends in the price series.
- **Moving Average-50 (MA50):** 50-day average of daily closes, was calculated for the last day of each month to capture medium-term trends in the price range.
- **Momentum:** Difference between the current close and the 63 trading days earlier close (approximately three months), providing trend speed and direction of price change.
- **Volatility:** 20-day moving standard deviation, calculated for the last day of each month to measure short-term changes in price series.
- **Relative Strength Index (RSI):** 14-day average of profit and loss, was calculated for the last day of each month to signal overbought or oversold areas.

Lagged Returns

Since Random Forest, XGBoost, and SVR models cannot capture temporal dependence, 1-, 2-, and 3-month lagged values were derived from the monthly return series to teach models the temporal dependence of returns. These variables aim to capture the effects of past monthly returns on future monthly returns.

Scaling

Since SVR and LSTM are scale-sensitive, all variables were standardized with the StandardScaler method (mean=0 and std=1) to make models learn more balanced and faster to ensure variables of different sizes have an equal impact on the model training process.

The feature set, combined with technical indicators and lagged returns, allows RF, XGBoost, and SVR to capture stock characteristics and time effects. Therefore, two models were developed for these algorithms: one with technical indicators and lagged returns, and the other with only indicators. Because LSTMs inherently capture time dependence, the feature set with lagged returns was not used.

3.4 Data Preprocessing

Data preprocessing plays a crucial role in ensuring the modeling process produces reliable and valid results. In this study, preprocessing was carried out in three main stages: managing missing data, dividing the dataset into time series, and preventing data leakage risks.

Missing Values

Calculating technical indicators and deriving the monthly return column created missing values. Since missing data negatively impacts model performance, NaN values were removed from the data set after feature engineering process to help models learn and preserve the integrity of the time-series.

Data Division: Train-Validation-Test Set

The dataset was divided into three subsets for training (70%), hyperparameter optimization (validation) (15%) and testing (15%) sets.

- The models were trained using the training set.
- The validation set was used to optimize hyperparameters and reduce the risk of overfitting.
- The test set was used to measure the model's performance on data it had never seen before and was used for final evaluation.

The data was divided chronologically since time series data flows and directions from the past to the future. This prevented future information from leaking into past modeling.

Data Leakage Precautions

To prevent data leakage, data was separated chronologically, technical indicators were calculated from historical data, and scaling was applied only to the training set. This prevented information leakage from the test data into the training process. Furthermore, using the TimeSeriesSplit method in hyperparameter optimization also prevented forward-looking leakage.

These preprocessing steps ensured that the dataset retained its statistical significance and allowed the modeling process to proceed without any risk of bias or overfitting. Thus, the resulting predictive performance represents a high level of real-world applicability, demonstrating the true performance of the models.

3.5 Sectoral Classification

DJIA companies were divided into 10 categories: technology, financials, consumer discretionary, consumer staples, industrials, healthcare, energy, communication services, materials, and insurance (Table 2). However, due to insufficient stock numbers in all categories, sector comparisons were limited to technology, financials, consumer discretionary, industrials, and healthcare.

Table 2: DJIA Sectors and Companies

Technology	Apple, Microsoft, IBM, Cisco, Nvidia, Salesforce
Financials	JPMorgan Chase, Goldman Sachs, American Express, Visa
Consumer Discretionary	Amazon, McDonald's, Home Depot, Nike, Disney
Consumer Staples	Procter & Gamble, Coca-Cola, Walmart
Industrials	3M, Boeing, Caterpillar, Honeywell
Healthcare	Johnson & Johnson, Merck, Amgen, UnitedHealth Group
Energy	Chevron
Communication Services	Verizon, Disney
Materials	Sherwin-Williams
Insurance	Travelers

3.6 Modeling Techniques

In this study, monthly returns were predicted with ML models to capture nonlinear and long-term relationships. All models except LSTM were tested in two scenarios:

1. Monthly return prediction with using close price and technical indicators.
2. Montly return prediction with using lagged (1,2,3 month) returns, technical indicators and close price.

The reason for not using lagged-returns for LSTM was, the model is capable of tracking past history data and can capture the relationship between current and prior returns.

3.6.1 Random Forest (RF)

Random Forest is an ensemble method created by combining multiple decision trees and is particularly effective at understanding nonlinear relationships among variables in multidimensional, complex datasets. The model's ability to generate variable importance rankings, coupled with the ability of multiple trees to learn different complexities, is expected to provide additional benefits for assessing the predictive power of technical indicators and delayed returns.

Hyperparameters (n_estimators, max_depth, min_samples_split, min_samples_leaf, max_features) were optimized with RandomizedSearchCV and TimeSeriesSplit. The training and validation sets were combined, the model was retrained with the new parameters, and the evaluation was performed on the test set.

3.6.2 XGBoost

This optimized model of decision trees can reduce the risk of overfitting thanks to its speed and editing features.

For hyperparameter optimization, the parameters n_estimators, max_depth, learning_rate, subsample, colsample_bytree, and min_child_weight were tested. Cross-validation was performed with TimeSeriesSplit, and the model was trained using the best parameters from the training and validation sets, and its performance was evaluated on the test set.

3.6.3 Support Vector Regression (SVR)

SVR is a method that is expected to be suitable for use in financial data sets due to its ability to model nonlinear relationships in multidimensional space through kernel functions.

In SVR models, before training, all variables were rescaled with StandardScaler to have a mean of 0 and a standard deviation of 1. Kernel, C, epsilon, and gamma parameters were optimized for hyperparameter optimization.

3.6.4 Long Short-Term Memory (LSTM)

LSTM is an RNN architecture that can learn long-term dependencies and evaluate historical data importance with input/output/forget gates. It has been used due to its potential to understand seasonality, trends, and historical data.

Model inputs were generated from consecutive periods with specific lookbacks, using only monthly closing prices and indicators. LSTM layers consisted of a dropout layer and a single-output density layer to prevent overfitting.

To prevent overfitting due to small dataset, early stopping was implemented, and training was completed when improvement stopped. Batch size, epoch, hidden unit, and dropout rate were also optimized.

3.7 Model Evaluation

To assess model performance, both regression-focused error metrics and directional accuracy, which is particularly important in financial forecasting were used. These metrics demonstrated the models' not only forecast errors but also directional accuracy metric which has crucial role for return in investment decisions.

3.7.1 Root Mean Squared Error (RMSE)

RMSE is the root-mean square of the differences between the predicted values and the true values.

$$RMSE = \sqrt{\frac{1}{n}\sum_{\{i=1\}}^{\{n\}} (\widehat{\{y\}}_i - y_i)^2}$$

Because it penalizes large errors more severely, it has been used to measure how sensitive the model is to extreme deviations, especially in financial time series with sudden price fluctuations.

3.7.2 Mean Absolute Error (MAE)

MAE is the average of the absolute values of the forecast errors.

$$MAE = \frac{1}{n}\sum_{\{i=1\}}^{\{n\}} |\widehat{\{y\}}_i - y_i|$$

It is easy to interpret and less affected by outliers than RMSE. Therefore, it is used as an additional measure to indicate the average magnitude of the model's overall error.

3.7.3 Directional Accuracy (DA)

Directional Accuracy measures the rate at which the model correctly predicts the direction of return (positive or negative).

$$DA = \frac{Correctly\ predicted\ directions}{All\ predictions}$$

Directional success is a leading method for evaluating investment strategies and has a direct impact on profitability. Models that predict not only numerical returns but also the direction of increase or decline are considered to evaluate the model's performance.

4. RESULTS

This section presents empirical results from models implemented for DJIA stocks. Following the overall performance comparison, the role of technical indicators will be presented. Afterwards, stock-level characteristics and model performances will be evaluated. Following that, most and least accurate forecasts will be shared through stock and metric-based analyses. The section ends with an evaluation of sector-specific results.

4.1 Overall Performance Comparison

In this section, the performances of 30 stocks included in the DJIA index were compared by taking the average of the test results of each model (Table 3).

Table 3: Overall Model Performances

General Results			
Random Forest	**RMSE**	**MAE**	**DA**
Technical Indicators	**0,077**	0,081	0,506
Technical Indicators with Lagged Returns	**0,077**	0,081	0,506
XGBoost	**RMSE**	**MAE**	**DA**
Technical Indicators	0,080	0,065	0,494
Technical Indicators with Lagged Returns	0,094	**0,063**	0,508
SVR	**RMSE**	**MAE**	**DA**
Technical Indicators	0,084	0,068	0,504
Technical Indicators with Lagged Returns	0,080	0,064	0,483
LSTM	**RMSE**	**MAE**	**DA**
Technical Indicators	0,124	0,113	**0,521**

The lowest RMSE value (0,077) was obtained in the Random Forest models both trained with technical indicators and indicators with lagged returns. This result demonstrates the RF algorithm's tree-based structure's ability to model complex relationships and capture noise in highly variable financial data.

The lowest MAE value (0.063) was observed in the XGBoost model trained with lag returns and technical indicators. This suggests that XGBoost can reduce absolute error by evaluating lagged values in some scenarios. However, the increase in RMSE indicates that the model cannot reduce some large errors but can reduce the overall average error.

The highest average directional accuracy was observed in the LSTM model, 52.1%. This indicates that LSTM is sensitive to directional patterns and seasonality effects in time series data, while high RMSE and MAE values indicate that it has difficulty predicting monthly returns.

While adding lagged returns generally improved RMSE for SVM, it did not significantly increase overall performance for most models. This suggests that adding lagged returns is not equally beneficial for all stocks and models.

The literature indicates that RF and XGBoost models are prominent in yield prediction (Gu et al., 2020), and this study confirms this with low RMSE and MAE results. Furthermore, the high success rate of LSTM in direction prediction (Fischer and Krauss, 2018) is confirmed by the LSTM model achieving the highest DA. The findings highlight that different models achieve strong performance in different domains, depending on the targeted metrics and data features.

4.2 Role of Technical Indicators

Technical indicators were the main components in all models used in this study. The findings indicated that technical indicators made a significant contribution to directional prediction and error metrics. Because these indicators convey the behavioral movements of monthly stock returns, they provided meaningful information about stock behavior in model estimation.

In the initial phases, the absence of technical indicators showed that close prices alone did not provide much meaning for stock returns. These indicators reflected the behavioral movements of the monthly stock returns, providing meaningful signals. However, the contribution of these indicators should be evaluated holistically, not solely in terms of their individual importance rankings. Feature importance analyses conducted in the initial phases of the study highlighted different indicators for different stocks in their importance rankings. However, when low-importance indicators were removed from the models, model performance

generally decreased. This demonstrates that the technical indicator set produced is holistically meaningful; that is, the model requires the entire set rather than relying solely on a few indicators.

Overall, this study shows that technical indicators are indispensable forecasting tools, enhancing prediction performance by carrying related information about monthly stock returns. Moreover, their contribution should be assessed holistically rather than feature importance rankings, meaning that even low-ranked indicators carry relevant information about stock returns.

4.3 Stock-Level Conditions and Model Performance

This section examines six stocks from among thirty that exhibit distinct characteristics and remarkable model performance. The aim is to examine the impact of trend structure, volatility, and characteristic differences on model performance through examples.

Stocks with low returns and standard deviations are considered stable and defensive, those with high returns and standard deviations are considered high volatility, and those in between are considered medium volatility and trending stocks. The results for the six stocks are below.

Table 4: Mean and Std. of Monthly Return for Selected Stocks for Classification

Stock	Mean Monthly Return	Std. Monthly Return
JNJ	0,007999	0,046489
PG	0,007765	0,053527
NVDA	0,047042	0,174128
BA	0,011140	0,095137
MSFT	0,015775	0,080076
CAT	0,019280	0,093291

Stable and Defensive Stocks

The LSTM model's strong directional prediction performance was generally noteworthy in low-volatility and stable stocks.

PG, a defensive and low-volatility stock, is an example where LSTM is particularly strong (Table 5). The LSTM model produced one of the highest values in the study, at 65.9%.

Table 5: PG Model Results

PG			
Random Forest	**RMSE**	**MAE**	**DA**
Technical Indicators	0,052	0,041	0,400
Technical Indicators with Lagged Returns	0,050	0,039	0,444
XGBoost	**RMSE**	**MAE**	**DA**
Technical Indicators	0,050	0,040	0,555
Technical Indicators with Lagged Returns	0,051	0,039	0,600
SVR	**RMSE**	**MAE**	**DA**
Technical Indicators	0,053	0,042	0,377
Technical Indicators with Lagged Returns	0,051	0,040	0,422
LSTM	**RMSE**	**MAE**	**DA**
Technical Indicators	0,066	0,052	0,659

Likewise, adding lagged returns improved the performance of other models. Similarly, JNJ, a defensive and low-volatility stock (Table 6), also demonstrated strong LSTM performance, and the addition of lagged returns increased the model's directional prediction percentage.

Table 6: JNJ Model Results

JNJ			
Random Forest	**RMSE**	**MAE**	**DA**
Technical Indicators	0,045	0,037	0,533
Technical Indicators with Lagged Returns	0,044	0,036	0,622
XGBoost	**RMSE**	**MAE**	**DA**
Technical Indicators	0,044	0,036	0,555
Technical Indicators with Lagged Returns	0,044	0,035	0,555
SVR	**RMSE**	**MAE**	**DA**
Technical Indicators	0,049	0,039	0,488
Technical Indicators with Lagged Returns	0,044	0,035	0,600
LSTM	**RMSE**	**MAE**	**DA**
Technical Indicators	0,047	0,038	0,590

In general, these two examples demonstrate that lagged returns increase directional prediction accuracy in defensive, low-volatile companies, and highlight the success of LSTM.

High Volatility Stocks

NVDA, a high-volatility technology stock, achieved 54.5% directional accuracy with LSTM, but lagged-return SVR, RF, and XGBoost performed better (Table 7). This suggests that LSTM is sensitive to high-volatility shocks, and tree- and kernel-based algorithms perform better in this environment.

Table 7: NVDA Model Results

NVDA			
Random Forest	**RMSE**	**MAE**	**DA**
Technical Indicators	0,161	0,135	0,577
Technical Indicators with Lagged Returns	0,162	0,133	0,622
XGBoost	**RMSE**	**MAE**	**DA**
Technical Indicators	0,158	0,132	0,644
Technical Indicators with Lagged Returns	0,158	0,132	0,622
SVR	**RMSE**	**MAE**	**DA**
Technical Indicators	0,175	0,145	0,355
Technical Indicators with Lagged Returns	0,163	0,134	0,644
LSTM	**RMSE**	**MAE**	**DA**
Technical Indicators	0,218	0,176	0,545

Similarly, for a high-volatile stock like BA, LSTM achieved only a 45.4% accuracy in predicting direction, while the SVR model achieved a 55.5% accuracy with lagged returns (Table 8).

Table 8: BA Model Results

BA			
Random Forest	**RMSE**	**MAE**	**DA**
Technical Indicators	0,119	0,096	0,466
Technical Indicators with Lagged Returns	0,122	0,100	0,400
XGBoost	**RMSE**	**MAE**	**DA**
Technical Indicators	0,110	0,087	0,466
Technical Indicators with Lagged Returns	0,110	0,086	0,488
SVR	**RMSE**	**MAE**	**DA**
Technical Indicators	0,246	0,192	0,466
Technical Indicators with Lagged Returns	0,167	0,127	0,555
LSTM	**RMSE**	**MAE**	**DA**
Technical Indicators	0,110	0,090	0,454

These two examples suggest that LSTM performs worse in high-volatility stocks, while models like SVR and XGBoost perform better with lagged returns.

Medium Volatility and Trending Stocks

CAT and MSFT stocks are not as volatile as high-volatility stocks, but they exhibit more pronounced trends than defensive stocks.

CAT achieved a DA of 62.2% using only technical indicators in the XGBoost model. The addition of lagged returns reduced both SVR and XGBoost performance (Table 9).

Table 9: CAT Model Results

CAT			
Random Forest	**RMSE**	**MAE**	**DA**
Technical Indicators	0,096	0,077	0,555
Technical Indicators with Lagged Returns	0,097	0,075	0,555
XGBoost	**RMSE**	**MAE**	**DA**
Technical Indicators	0,096	0,076	0,622
Technical Indicators with Lagged Returns	0,096	0,075	0,533
SVR	**RMSE**	**MAE**	**DA**
Technical Indicators	0,099	0,079	0,600
Technical Indicators with Lagged Returns	0,097	0,078	0,533
LSTM	**RMSE**	**MAE**	**DA**
Technical Indicators	0,129	0,101	0,568

Likewise, MSFT achieved a DA of 61.3% using the LSTM model. However, LSTM's RMSE and MAE results for both stocks remained higher than those in the other models (Table 10).

Table 10: MSFT Model Results

MSFT			
Random Forest	**RMSE**	**MAE**	**DA**
Technical Indicators	0,070	0,057	0,577
Technical Indicators with Lagged Returns	0,073	0,058	0,555
XGBoost	**RMSE**	**MAE**	**DA**
Technical Indicators	0,075	0,059	0,555
Technical Indicators with Lagged Returns	0,079	0,062	0,555
SVR	**RMSE**	**MAE**	**DA**
Technical Indicators	0,087	0,070	0,555
Technical Indicators with Lagged Returns	0,077	0,062	0,555
LSTM	**RMSE**	**MAE**	**DA**
Technical Indicators	0,091	0,070	0,613

In general, lagged returns and time dependence (LSTM) increase DA in moderately volatile stocks, while they tend to increase RMSE and MAE values. However, these RMSE and MAE values may vary depending on the model and stocks.

Stock-based analyses show that lagged returns do not have the same effect on all stocks but rather improve performance in some. While LSTM performed well in predicting direction at low volatility, other models performed better with lagged returns at high volatility. Overall, LSTM performed better in predicting direction, while SVR, RF, and XGBoost were superior in terms of RMSE and MAE. The results confirm that model selection varies depending on stock structure, volatility, and trend dynamics, and no single model can be superior for all stocks. For more results, see Appendix A.

In summary, the results show that technical indicators and lagged returns vary depending on stock conditions. In this context, the next section will demonstrate which stocks and metrics stand out and examine differences in model performance based on stock characteristics.

4.4 Stock and Metric Comparisons

This study demonstrates that the models do not perform equally well across DJIA companies based on their stock-based analyses. While some stocks achieved good DA and RMSE performance, other stocks exhibited limited model performance.

4.4.1 Best Performing Stocks for Predictions

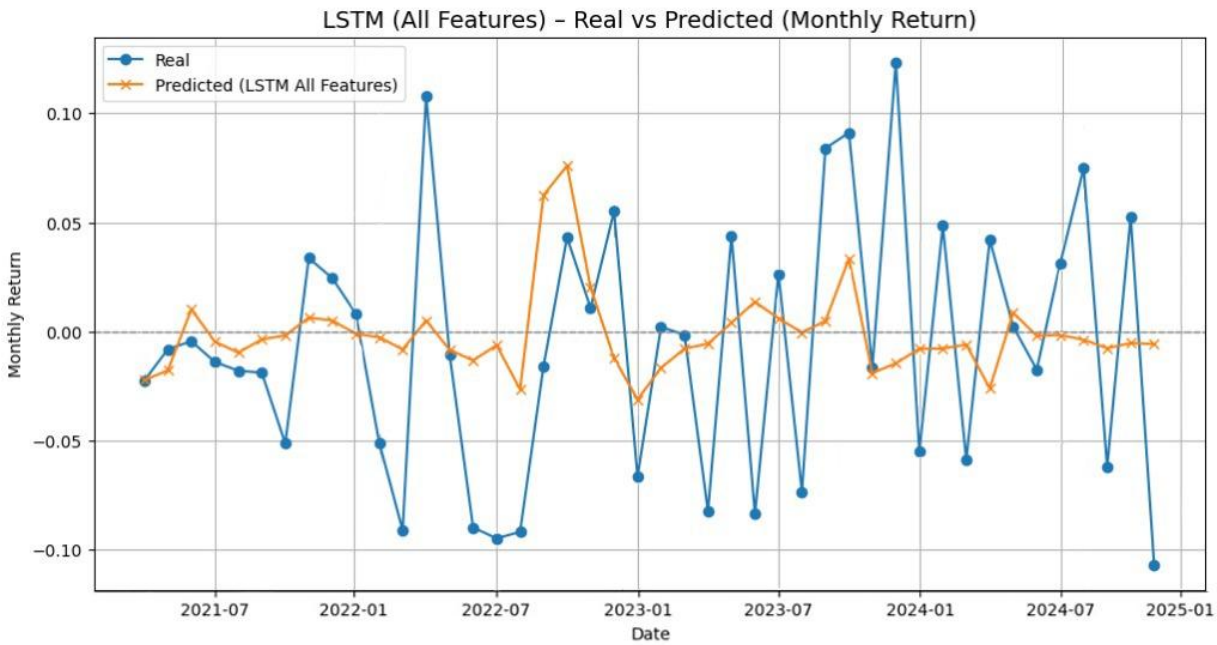

Figure 1: VZ LSTM Actual vs. Predicted Graph

The stocks with the highest directional accuracy were Verizon, P&G, and NVIDIA. Verizon achieved 72.3% accuracy with the LSTM model (Appendix Table A29). As seen in Figure 1, the model generally captured directional changes, but the predictions were more horizontal than the actual values. This suggests that while LSTM captures trends, but struggles to predict returns.

Likewise, P&G similarly achieved a DA of 65.9% with the LSTM model (Table – 5), demonstrating the directional success of the LSTM model in less volatile stocks like Verizon.

NVIDIA achieved a DA success rate of 64.4% with XGBoost and SVR lagged returns models, demonstrating that strong direction prediction can be made for stocks with high volatility by selecting the appropriate model (Table 6).

Secondly, stocks with the least error in terms of RMSE were Johnson & Johnson, Coca-Cola Company and McDonald's. It was striking that all three companies are defensively operating companies.

J&J achieved RMSEs in the 0.044-0.045 range in the RF and XGBoost models, suggesting accurate monthly return predictions in the low-volatile healthcare sector (Table – 4). The fact that some predictions directly matched the actual values shed light on the model's capability. While directional accuracy remained moderate, the low error metrics demonstrate that tree-based models can produce stable predictions (Figure 2).

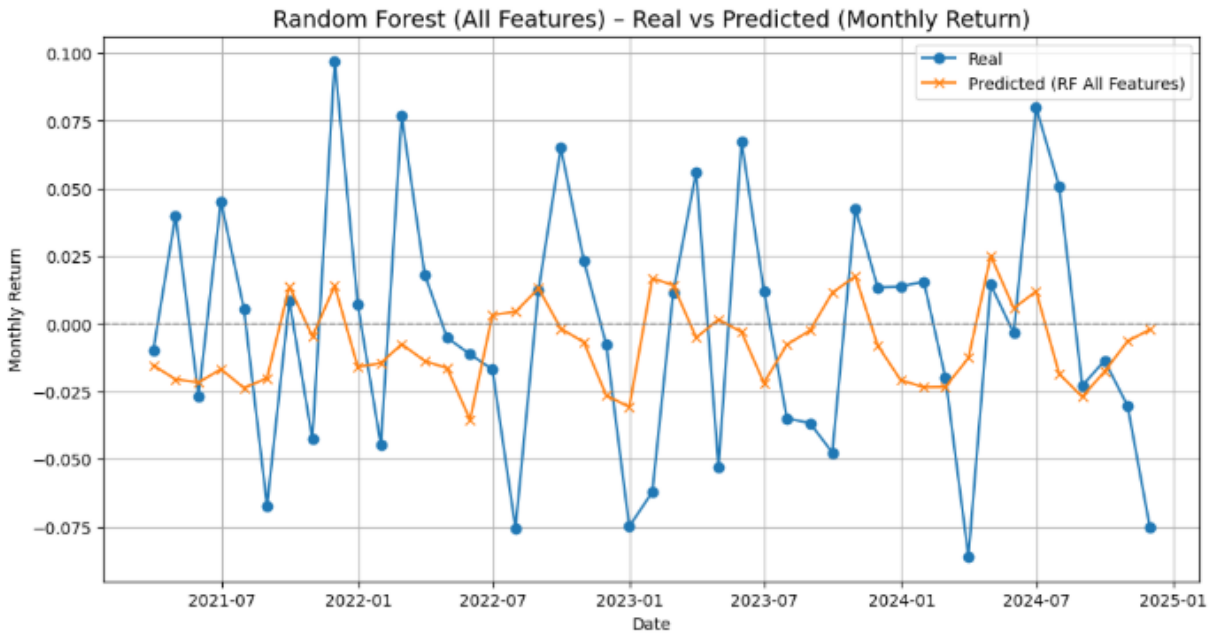

Figure 2: JNJ RF (All Features) Actual vs. Predicted Graph

Similarly, Coca-Cola and McDonald's stood out with RMSEs of 0.050-0.052, suggesting that upward-trending stocks are more likely to forecast better monthly returns. For the results, see Appendix A.

As a result, while the error metrics for stocks in defensive sectors were low, a high degree of directional accuracy was observed in low-volatility stocks where clear patterns are seen.

4.4.2 Worst Performing Stocks for Predictions

The lowest DA performance was observed in TRV, GS and V stocks.

TRV achieved 38.6% accuracy in the LSTM model, while SVR and RF achieved 40% and 37.7%. While the use of technical indicators increased this accuracy, overall forecasting performance did not satisfy.

Likewise, GS achieved 36.3% accuracy in the LSTM model, and RMSE of 0.263. Similarly, Visa stock obtained a similar result, with SVR achieving 33.3% and average results in other models. The fact that these two stocks are financial companies suggests that multidimensional factors such as interest rate decisions, crises, global payment trends, and fintech competition negatively impact directional forecasts.

Boeing's poor performance under high volatility caused forecast bias. While BA produced the highest RMSE in the SVR model at 0.236, lagged returns partially reduced the error, but accuracy remained limited. This suggests that forecasting power may be low in macroeconomically and geopolitically risky sectors (Figure - 3).

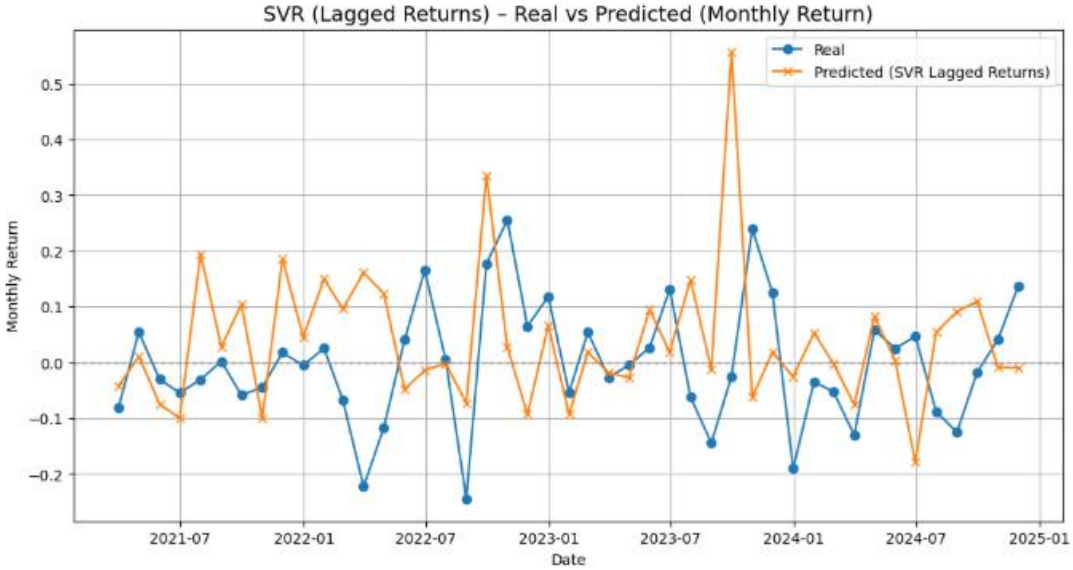

Figure 3: BA SVR (All Features) Actual vs. Predicted Graph

Overall, RMSE values of highly volatile stocks indicated poor performance. On the same hand, DA decreased in sectors sensitive to macroeconomic factors, such as finance, and those affected by sudden shocks. However, in addition to sector-based analyses, stocks with different characteristics can yield different results in different models.

4.4.3 Metric-Based Assessment

These findings offer two generalizations for these metrics:

1. RMSE and MAE were metrics that increase as volatility increases. High volatility hindered the model's forecast, causing deviations in predictions.
2. Directional accuracy, particularly in LSTM, was higher for stocks with medium-to-low volatility. DA was higher for stocks that were moving at support-resistance levels and did not experience extreme movements.

4.5 Sector-Level Results

A sector-level study revealed that models responded differently across sectors. These findings suggest that different sectors exhibit significant differences (Table 11).

Table 11: Overall Sector Performances

Sector	Average RMSE	Average MAE	Average DA
Technology	0,099	0,080	0,537
Financials	0,085	0,069	0,462
Consumer Discretionary	0,101	0,123	0,511
Industrials	0,098	0,078	0,496
Healthcare	0,064	0,051	0,503

Firstly, the healthcare sector produced the lowest error rates compared to other sectors. This result suggests that stock returns in the healthcare sector are relatively more predictable. The absence of large shocks and sudden fluctuations in price movements increased the predictability of the sector's monthly returns. The DA ratio of 50.3% may be due to fact that prices mostly move in small increments. While it is easy to detect general trends, distinguishing them in the right direction remains challenging for models.

In contrast, the consumer discretionary sector produced the highest error metrics with second highest directional accuracy. This sector's sensitivity to macroeconomic conditions and fluctuations in consumer demand reduced predictability. Exposure to periodic high volatility

during long-term upward trends reduced predictability, but at the same time could help with predicting direction of monthly returns.

Likewise, technology sector performed averagely in terms of monthly return forecasting, with the highest DA score. These findings demonstrate that the models' strong directional predictions stem from the sector's strong trend components and dynamics driven by investor expectations. Furthermore, high volatility, just as it did in the consumer discretionary sector, also reduced predictability here, but increased directional prediction.

While price predictions in the financials were average, they exhibited the worst directional accuracy. As mentioned earlier, this may make directional predictions for financial stocks difficult due to multidimensional factors. For detailed sector-model results, see Appendix B.

The results demonstrate that ML methods, despite their limitations, offer significant predictive performance, with no single model outperforming all scenarios. Tree-based RF and XGBoost models yielded lower RMSE and MAE values, while LSTM stood out for its directional prediction performance on low-volatility and trending stocks. Furthermore, while technical indicators have varying importance for each stock, the model's performance deteriorated when low-importance indicators were removed during model development, demonstrating the overall significance of the indicators. Looking at stock-level characteristics, LSTM excelled at trend-capturing performance on defensive low-volatility stocks, while tree- and kernel-based models performed better on high-volatility stocks. At the sectoral level, the healthcare sector achieved the lowest error metrics, while the technology sector achieved the highest directional accuracy.

5. DISCUSSION

The results of this study, compared to literature, share both similarities and differences. In overall, ML methods applied for DJIA stocks produced low predictive power and failed to identify an absolute accurate model. In contradiction to that, they did highlight certain patterns in certain scenarios and produced meaningful directional prediction. This suggests that ML methods can produce small but significant predictions rather than absolute accuracy, as emphasized in the literature (Gu et al., 2020).

While each model stood out for stocks under different conditions, the RF and XGBoost models produced consistent results, demonstrating that tree-based methods can excel in financial time series (Freyberger et al., 2020). In contrast, SVM was generally weaker and experienced a performance loss in complex data. LSTM, on the other hand, stood out with its high directional prediction accuracy in certain stocks, but it produced unstable results. This demonstrates the prominence of different models for different sectors.

The use of a combination of technical indicators strengthened model performance. While different indicators were more important for different stocks, it was observed that model performance decreased when these indicators were removed. This finding demonstrates the holistic importance of the indicator set used. Similarly, it also demonstrates that indicator importance varies depending on market structure.

Models using lagged returns also improved performance on various stocks, but they failed to provide absolute improvement. This confirms that the success of ML models varies depending on the market context (Choi et al., 2023).

While the results do not invalidate Fama's (1970) strong form hypothesis in the context of EMH, they do question the absolute validity of the weak form. The high directional predictions obtained for some stocks indicate that meaningful information can be extracted from past prices.

Considering the study's limitations, the study's inclusion of only DJIA stocks reduced generalization capacity. Furthermore, the limited use of indicators reduced forecast performance. Moreover, the fact that there are not equal numbers of companies in each sector of the DJIA stocks caused not all sectors to be included in the study. Finally, the small size of the dataset due

to monthly processing and the inability to provide sufficient data, limits the learning capabilities of the models, especially LSTM.

In summary, this study demonstrates that ML methods have predictive power in DJIA stocks under certain conditions and sectors. While the RF and XGBoost models demonstrated more balanced performance, LSTM stood out with its strong directional predictions. This study, which identifies the holistic importance of technical indicators and sectoral differences, generally aligns with the literature, provides model comparisons, and offers new contributions for Dow Jones Industrial Average Index.

6. CONCLUSION AND FUTURE WORK

This study aimed to predict monthly returns with ML models for the stocks in the DJIA. Findings indicated that ML methods offer a limited but significant predictive power, which is not entirely random as EMH indicates.

Random Forest and XGBoost models stood out as the most successful models in error metrics, while LSTM produced particularly strong directional predictions for some stocks. While different indicators were more important for different stocks, removing less important indicators reduced model performance. This demonstrated that the indicators positively impacted the model's overall performance. Lagged return variables exhibited a complex effect that varied by stock.

In sector analyses, the healthcare sector was the most predictable sector with lower error metrics, while the technology sector was observed to be more predictable in direction, and finance sector performed relatively poorly. While high-volatility stocks had higher error metrics, the LSTM model provided a relative advantage in directional prediction.

These findings are consistent with results in the literature, demonstrating the importance of technical indicators and the decisive role of sector and market conditions in forecasting stocks. While academically, they demonstrate the potential and limitations of ML in financial time series, from a practical perspective, they offer investors significant insights, particularly regarding directional prediction.

This study could produce more consistent results by expanding the dataset to include daily or weekly data. Using hybrid models and trend indicators such as Bollinger Bands could improve model performance and produce more accurate monthly return predictions.

In conclusion, this study demonstrates that machine learning can provide meaningful but limited forecasts for DJIA stocks. The findings suggest that model performance may vary depending on sector and stock differences, with some sectors performing better. Future research with proposed methodologies can contribute to this field both academically and practically.

REFERENCES

- Adebiyi, A. A., Adewumi, A. O. & Ayo, C. K. (2014). 'Stock price prediction using the ARIMA model', *Proceedings of the 2014 UKSim-AMSS 16th International Conference on Computer Modelling and Simulation*, Cambridge, UK. 26–28 March 2014. IEEE. pp. 106–112. https://doi.org/10.1109/UKSim.2014.67.
- Althelaya, K. A., El-Alfy, E.-S. M. & Mohammed, S. (2018). 'Evaluation of bidirectional LSTM for short- and long-term stock market prediction', *Proceedings of the 2018 9th International Conference on Information and Communication Systems (ICICS)*, Irbid, Jordan. 3–5 April 2018. IEEE. pp. 151–156. https://doi.org/10.1109/IACS.2018.8355458.
- Avramov, D., Cheng, S. & Metzker, L. (2023). 'Machine learning vs. economic restrictions: Evidence from stock return predictability', *Management Science*, 69(5), pp. 2587–2619. https://doi.org/10.1287/mnsc.2022.4449.
- Ballings, M., Van den Poel, D., Hespeels, N. & Gryp, R. (2015). 'Evaluating multiple classifiers for stock price direction prediction', *Expert Systems with Applications*, 42(20), pp. 7046–7056. https://doi.org/10.1016/j.eswa.2015.05.013.
- Box, G. E. P. & Jenkins, G. M. (1976). *Time series analysis: forecasting and control*. Revised edn. San Francisco: Holden-Day.
- Chen, Y. & Hao, Y. (2017). 'A feature weighted support vector machine and k nearest neighbor algorithm for stock market indices prediction', *Expert Systems with Applications*, 80, pp. 340–355. https://doi.org/10.1016/j.eswa.2017.02.044.
- Choi, D., Jiang, W. & Zhang, C. (2023). 'Alpha Go everywhere: Machine learning and international stock returns', *Review of Finance*, (in press). Available at: https://ssrn.com/abstract=3489679 (Accessed: 6 July 2025).
- Fama, E. F. (1970). 'Efficient capital markets: A review of theory and empirical work', *The Journal of Finance*, 25(2), pp. 383–417. https://doi.org/10.2307/2325486.
- Fischer, T. & Krauss, C. (2018). 'Deep learning with long short-term memory networks for financial market predictions', *European Journal of Operational Research*, 270(2), pp. 654–669. https://doi.org/10.1016/j.ejor.2017.11.054.
- Freyberger, J., Neuhierl, A. & Weber, M. (2020). 'Dissecting characteristics nonparametrically', *The Review of Financial Studies*, 33(5), pp. 2326–2377. https://doi.org/10.1093/rfs/hhz123.
- Gencay, R. & Stengos, T. (1998). 'Moving average rules, volume and the predictability of security returns with feed-forward networks', *Journal of Forecasting*, 17(5–6), pp. 401–414. https://doi.org/10.1002/(SICI)1099-131X(1998090)17:5/6<401::AID-FOR704>3.0.CO;2-C.
- Gu, S., Kelly, B. & Xiu, D. (2020). 'Empirical asset pricing via machine learning', *The Review of Financial Studies*, 33(5), pp. 2223–2273. https://doi.org/10.1093/rfs/hhaa009.

- Han, Y., He, A., Rapach, D. E. & Zhou, G. (2018). 'What firm characteristics drive US stock returns?', *Review of Finance*, 23(4), pp. 637–682.
- Hansson, M. (2017). *On stock return prediction with LSTM networks*. MSc Thesis. Department of Economics, Lund University, Sweden.
- Kim, H. Y. & Won, C. H. (2018). 'Forecasting the volatility of stock price index: A hybrid model integrating LSTM with multiple GARCH-type models', *Expert Systems with Applications*, 103, pp. 25–37. https://doi.org/10.1016/j.eswa.2018.03.002.
- Kumbure, M. M., Lohrmann, C., Luukka, P. & Porras, J. (2022). 'Machine learning techniques and data for stock market forecasting: A literature review', *Expert Systems with Applications*, 197, 116659. https://doi.org/10.1016/j.eswa.2022.116659.
- Leippold, M., Wang, Q. & Zhou, W. (2022). 'Machine learning in the Chinese stock market', *Journal of Financial Economics*, 145(1), pp. 64–82. https://doi.org/10.1016/j.jfineco.2021.08.017.
- Sun, G. & Deng, S. (2025). 'Financial time series forecasting: A comparison between traditional methods and AI-driven techniques', *Journal of Computer, Signal, and System Research*, 2(2), pp. 1–8. https://doi.org/10.71222/339b9812.
- Tsay, R. S. (2010). *Analysis of financial time series*. 3rd edn. Hoboken, NJ: John Wiley & Sons.
- Uzzal, M. H. & Ślepaczuk, R. (2023). 'The performance of time series forecasting based on classical and machine learning methods for S&P 500 index', *Working Papers*, 2023-05. Faculty of Economic Sciences, University of Warsaw. Available at: https://ideas.repec.org/p/war/wpaper/2023-05.html (Accessed: 7 July 2025).
- Wang, W. (2024). 'Comparative analysis of ARIMA, random forest, and LSTM models for Mercedes-Benz stock price prediction', *Advances in Economics, Management and Political Sciences*, 134, pp. 1–8. https://doi.org/10.54254/2754-1169/2024.18718.
- Yadav, A., Jha, C. K. & Sharan, A. (2020). 'Optimizing LSTM for time series prediction in Indian stock market', *Procedia Computer Science*, 167, pp. 2091–2100. https://doi.org/10.1016/j.procs.2020.03.257.

APPENDIX

A. Stock-Based Model Results

Table A1: AAPL Model Results

AAPL			
Random Forest	**RMSE**	**MAE**	**DA**
Technical Indicators	0,077	0,062	0,555
Technical Indicators with Lagged Returns	0,074	0,061	0,600
XGBoost	**RMSE**	**MAE**	**DA**
Technical Indicators	0,080	0,064	0,600
Technical Indicators with Lagged Returns	0,078	0,062	0,555
SVR	**RMSE**	**MAE**	**DA**
Technical Indicators	0,076	0,064	0,444
Technical Indicators with Lagged Returns	0,073	0,062	0,577
LSTM	**RMSE**	**MAE**	**DA**
Technical Indicators	0,246	0,209	0,454

Table A2: AMGN Model Results

AMGN			
Random Forest	**RMSE**	**MAE**	**DA**
Technical Indicators	0,070	0,057	0,488
Technical Indicators with Lagged Returns	0,070	0,058	0,444
XGBoost	**RMSE**	**MAE**	**DA**
Technical Indicators	0,070	0,057	0,533
Technical Indicators with Lagged Returns	0,070	0,058	0,444
SVR	**RMSE**	**MAE**	**DA**
Technical Indicators	0,073	0,059	0,444
Technical Indicators with Lagged Returns	0,088	0,073	0,533
LSTM	**RMSE**	**MAE**	**DA**
Technical Indicators	0,087	0,071	0,522

Table A3: AMZN Model Results

AMZN			
Random Forest	**RMSE**	**MAE**	**DA**
Technical Indicators	0,095	0,073	0,600
Technical Indicators with Lagged Returns	0,097	0,073	0,600
XGBoost	**RMSE**	**MAE**	**DA**
Technical Indicators	0,095	0,072	0,600
Technical Indicators with Lagged Returns	0,098	0,075	0,600
SVR	**RMSE**	**MAE**	**DA**
Technical Indicators	0,100	0,078	0,577
Technical Indicators with Lagged Returns	0,096	0,074	0,577
LSTM	**RMSE**	**MAE**	**DA**
Technical Indicators	0,131	0,109	0,430

Table A4: AXP Model Results

AXP			
Random Forest	**RMSE**	**MAE**	**DA**
Technical Indicators	0,081	0,064	0,466
Technical Indicators with Lagged Returns	0,080	0,066	0,355
XGBoost	**RMSE**	**MAE**	**DA**
Technical Indicators	0,112	0,095	0,444
Technical Indicators with Lagged Returns	0,081	0,067	0,422
SVR	**RMSE**	**MAE**	**DA**
Technical Indicators	0,081	0,065	0,533
Technical Indicators with Lagged Returns	0,077	0,064	0,511
LSTM	**RMSE**	**MAE**	**DA**
Technical Indicators	0,1233	0,095	0,545

Table A5: BA Model Results

BA			
Random Forest	**RMSE**	**MAE**	**DA**
Technical Indicators	0,119	0,096	0,466
Technical Indicators with Lagged Returns	0,122	0,100	0,400
XGBoost	**RMSE**	**MAE**	**DA**
Technical Indicators	0,110	0,087	0,466
Technical Indicators with Lagged Returns	0,110	0,086	0,488
SVR	**RMSE**	**MAE**	**DA**
Technical Indicators	0,246	0,192	0,466
Technical Indicators with Lagged Returns	0,167	0,127	0,555
LSTM	**RMSE**	**MAE**	**DA**
Technical Indicators	0,110	0,090	0,454

Table A6: CAT Model Results

CAT			
Random Forest	**RMSE**	**MAE**	**DA**
Technical Indicators	0,096	0,077	0,555
Technical Indicators with Lagged Returns	0,097	0,075	0,555
XGBoost	**RMSE**	**MAE**	**DA**
Technical Indicators	0,096	0,076	0,622
Technical Indicators with Lagged Returns	0,096	0,075	0,533
SVR	**RMSE**	**MAE**	**DA**
Technical Indicators	0,099	0,079	0,600
Technical Indicators with Lagged Returns	0,097	0,078	0,533
LSTM	**RMSE**	**MAE**	**DA**
Technical Indicators	0,129	0,101	0,568

Table A7: CRM Model Results

CRM			
Random Forest	**RMSE**	**MAE**	**DA**
Technical Indicators	0,111	0,091	0,486
Technical Indicators with Lagged Returns	0,109	0,090	0,472
XGBoost	**RMSE**	**MAE**	**DA**
Technical Indicators	0,111	0,092	0,486
Technical Indicators with Lagged Returns	0,108	0,080	0,500
SVR	**RMSE**	**MAE**	**DA**
Technical Indicators	0,108	0,086	0,567
Technical Indicators with Lagged Returns	0,119	0,096	0,500
LSTM	**RMSE**	**MAE**	**DA**
Technical Indicators	0,107	0,085	0,638

Table A8: CSCO Model Results

CSCO			
Random Forest	**RMSE**	**MAE**	**DA**
Technical Indicators	0,065	0,054	0,488
Technical Indicators with Lagged Returns	0,064	0,052	0,577
XGBoost	**RMSE**	**MAE**	**DA**
Technical Indicators	0,063	0,051	0,555
Technical Indicators with Lagged Returns	0,063	0,050	0,600
SVR	**RMSE**	**MAE**	**DA**
Technical Indicators	0,066	0,053	0,511
Technical Indicators with Lagged Returns	0,064	0,051	0,533
LSTM	**RMSE**	**MAE**	**DA**
Technical Indicators	0,068	0,058	0,454

Table A9: CVX Model Results

CVX			
Random Forest	**RMSE**	**MAE**	**DA**
Technical Indicators	0,080	0,060	0,466
Technical Indicators with Lagged Returns	0,081	0,060	0,400
XGBoost	**RMSE**	**MAE**	**DA**
Technical Indicators	0,078	0,059	0,533
Technical Indicators with Lagged Returns	0,078	0,058	0,555
SVR	**RMSE**	**MAE**	**DA**
Technical Indicators	0,082	0,063	0,466
Technical Indicators with Lagged Returns	0,082	0,062	0,556
LSTM	**RMSE**	**MAE**	**DA**
Technical Indicators	0,138	0,109	0,477

Table A10: DIS Model Results

DIS			
Random Forest	**RMSE**	**MAE**	**DA**
Technical Indicators	0,098	0,077	0,511
Technical Indicators with Lagged Returns	0,100	0,080	0,444
XGBoost	**RMSE**	**MAE**	**DA**
Technical Indicators	0,098	0,077	0,466
Technical Indicators with Lagged Returns	0,105	0,082	0,466
SVR	**RMSE**	**MAE**	**DA**
Technical Indicators	0,108	0,082	0,511
Technical Indicators with Lagged Returns	0,099	0,080	0,444
LSTM	**RMSE**	**MAE**	**DA**
Technical Indicators	0,118	0,099	0,522

Table A11: GS Model Results

GS			
Random Forest	**RMSE**	**MAE**	**DA**
Technical Indicators	0,084	0,071	0,511
Technical Indicators with Lagged Returns	0,084	0,071	0,422
XGBoost	**RMSE**	**MAE**	**DA**
Technical Indicators	0,086	0,072	0,488
Technical Indicators with Lagged Returns	0,081	0,069	0,444
SVR	**RMSE**	**MAE**	**DA**
Technical Indicators	0,078	0,068	0,511
Technical Indicators with Lagged Returns	0,080	0,068	0,488
LSTM	**RMSE**	**MAE**	**DA**
Technical Indicators	0,263	0,204	0,363

Table A12: HD Model Results

HD			
Random Forest	**RMSE**	**MAE**	**DA**
Technical Indicators	0,075	0,620	0,511
Technical Indicators with Lagged Returns	0,074	0,615	0,555
XGBoost	**RMSE**	**MAE**	**DA**
Technical Indicators	0,076	0,063	0,466
Technical Indicators with Lagged Returns	0,076	0,062	0,577
SVR	**RMSE**	**MAE**	**DA**
Technical Indicators	0,095	0,076	0,577
Technical Indicators with Lagged Returns	0,068	0,055	0,577
LSTM	**RMSE**	**MAE**	**DA**
Technical Indicators	0,128	0,103	0,477

Table A13: HON Model Results

HON			
Random Forest	**RMSE**	**MAE**	**DA**
Technical Indicators	0,065	0,050	0,422
Technical Indicators with Lagged Returns	0,064	0,049	0,511
XGBoost	**RMSE**	**MAE**	**DA**
Technical Indicators	0,068	0,053	0,377
Technical Indicators with Lagged Returns	0,063	0,051	0,466
SVR	**RMSE**	**MAE**	**DA**
Technical Indicators	0,061	0,048	0,555
Technical Indicators with Lagged Returns	0,060	0,048	0,488
LSTM	**RMSE**	**MAE**	**DA**
Technical Indicators	0,075	0,062	0,450

Table A14: IBM Model Results

IBM			
Random Forest	**RMSE**	**MAE**	**DA**
Technical Indicators	0,065	0,052	0,555
Technical Indicators with Lagged Returns	0,067	0,054	0,444
XGBoost	**RMSE**	**MAE**	**DA**
Technical Indicators	0,065	0,053	0,488
Technical Indicators with Lagged Returns	0,066	0,054	0,377
SVR	**RMSE**	**MAE**	**DA**
Technical Indicators	0,068	0,057	0,555
Technical Indicators with Lagged Returns	0,066	0,053	0,466
LSTM	**RMSE**	**MAE**	**DA**
Technical Indicators	0,066	0,055	0,533

Table A15: JNJ Model Results

JNJ			
Random Forest	**RMSE**	**MAE**	**DA**
Technical Indicators	0,045	0,037	0,533
Technical Indicators with Lagged Returns	0,044	0,036	0,622
XGBoost	**RMSE**	**MAE**	**DA**
Technical Indicators	0,044	0,036	0,555
Technical Indicators with Lagged Returns	0,044	0,035	0,555
SVR	**RMSE**	**MAE**	**DA**
Technical Indicators	0,049	0,039	0,488
Technical Indicators with Lagged Returns	0,044	0,035	0,600
LSTM	**RMSE**	**MAE**	**DA**
Technical Indicators	0,047	0,038	0,590

Table A16: JPM Model Results

JPM			
Random Forest	**RMSE**	**MAE**	**DA**
Technical Indicators	0,070	0,059	0,533
Technical Indicators with Lagged Returns	0,073	0,061	0,466
XGBoost	**RMSE**	**MAE**	**DA**
Technical Indicators	0,095	0,076	0,511
Technical Indicators with Lagged Returns	0,086	0,070	0,444
SVR	**RMSE**	**MAE**	**DA**
Technical Indicators	0,077	0,063	0,4667
Technical Indicators with Lagged Returns	0,073	0,060	0,0488
LSTM	**RMSE**	**MAE**	**DA**
Technical Indicators	0,087	0,069	0,577

Table A17: KO Model Results

KO			
Random Forest	**RMSE**	**MAE**	**DA**
Technical Indicators	0,051	0,041	0,422
Technical Indicators with Lagged Returns	0,041	0,041	0,444
XGBoost	**RMSE**	**MAE**	**DA**
Technical Indicators	0,080	0,071	0,400
Technical Indicators with Lagged Returns	0,073	0,064	0,444
SVR	**RMSE**	**MAE**	**DA**
Technical Indicators	0,058	0,047	0,400
Technical Indicators with Lagged Returns	0,050	0,040	0,400
LSTM	**RMSE**	**MAE**	**DA**
Technical Indicators	0,507	0,039	0,568

Table A18: MCD Model Results

MCD			
Random Forest	**RMSE**	**MAE**	**DA**
Technical Indicators	0,053	0,039	0,488
Technical Indicators with Lagged Returns	0,051	0,038	0,555
XGBoost	**RMSE**	**MAE**	**DA**
Technical Indicators	0,052	0,038	0,555
Technical Indicators with Lagged Returns	0,506	0,038	0,511
SVR	**RMSE**	**MAE**	**DA**
Technical Indicators	0,052	0,038	0,533
Technical Indicators with Lagged Returns	0,050	0,037	0,511
LSTM	**RMSE**	**MAE**	**DA**
Technical Indicators	0,055	0,042	0,568

Table A19: MMM Model Results

MMM			
Random Forest	**RMSE**	**MAE**	**DA**
Technical Indicators	0,083	0,067	0,466
Technical Indicators with Lagged Returns	0,084	0,067	0,444
XGBoost	**RMSE**	**MAE**	**DA**
Technical Indicators	0,084	0,067	0,466
Technical Indicators with Lagged Returns	0,084	0,067	0,466
SVR	**RMSE**	**MAE**	**DA**
Technical Indicators	0,091	0,070	0,511
Technical Indicators with Lagged Returns	0,087	0,068	0,488
LSTM	**RMSE**	**MAE**	**DA**
Technical Indicators	0,089	0,072	0,522

Table A20: MRK Model Results

MRK			
Random Forest	**RMSE**	**MAE**	**DA**
Technical Indicators	0,067	0,049	0,577
Technical Indicators with Lagged Returns	0,069	0,051	0,577
XGBoost	**RMSE**	**MAE**	**DA**
Technical Indicators	0,070	0,052	0,555
Technical Indicators with Lagged Returns	0,069	0,050	0,577
SVR	**RMSE**	**MAE**	**DA**
Technical Indicators	0,069	0,052	0,577
Technical Indicators with Lagged Returns	0,067	0,050	0,533
LSTM	**RMSE**	**MAE**	**DA**
Technical Indicators	0,085	0,066	0,488

Table A21: MSFT Model Results

MSFT			
Random Forest	**RMSE**	**MAE**	**DA**
Technical Indicators	0,070	0,057	0,577
Technical Indicators with Lagged Returns	0,073	0,058	0,555
XGBoost	**RMSE**	**MAE**	**DA**
Technical Indicators	0,075	0,059	0,555
Technical Indicators with Lagged Returns	0,079	0,062	0,555
SVR	**RMSE**	**MAE**	**DA**
Technical Indicators	0,087	0,070	0,555
Technical Indicators with Lagged Returns	0,077	0,062	0,555
LSTM	**RMSE**	**MAE**	**DA**
Technical Indicators	0,091	0,070	0,613

Table A22: NKE Model Results

NKE			
Random Forest	**RMSE**	**MAE**	**DA**
Technical Indicators	0,097	0,077	0,511
Technical Indicators with Lagged Returns	0,096	0,076	0,533
XGBoost	**RMSE**	**MAE**	**DA**
Technical Indicators	0,097	0,077	0,051
Technical Indicators with Lagged Returns	0,096	0,077	0,488
SVR	**RMSE**	**MAE**	**DA**
Technical Indicators	0,097	0,079	0,466
Technical Indicators with Lagged Returns	0,094	0,075	0,466
LSTM	**RMSE**	**MAE**	**DA**
Technical Indicators	0,105	0,85	0,545

Table A23: NVDA Model Results

NVDA			
Random Forest	**RMSE**	**MAE**	**DA**
Technical Indicators	0,161	0,135	0,577
Technical Indicators with Lagged Returns	0,162	0,133	0,622
XGBoost	**RMSE**	**MAE**	**DA**
Technical Indicators	0,158	0,132	0,644
Technical Indicators with Lagged Returns	0,158	0,132	0,622
SVR	**RMSE**	**MAE**	**DA**
Technical Indicators	0,175	0,145	0,355
Technical Indicators with Lagged Returns	0,163	0,134	0,644
LSTM	**RMSE**	**MAE**	**DA**
Technical Indicators	0,218	0,176	0,545

Table A24: PG Model Results

PG			
Random Forest	**RMSE**	**MAE**	**DA**
Technical Indicators	0,052	0,041	0,400
Technical Indicators with Lagged Returns	0,050	0,039	0,444
XGBoost	**RMSE**	**MAE**	**DA**
Technical Indicators	0,050	0,040	0,555
Technical Indicators with Lagged Returns	0,051	0,039	0,60
SVR	**RMSE**	**MAE**	**DA**
Technical Indicators	0,053	0,042	0,377
Technical Indicators with Lagged Returns	0,051	0,040	0,422
LSTM	**RMSE**	**MAE**	**DA**
Technical Indicators	0,066	0,052	0,659

Table A25: SHW Model Results

SHW			
Random Forest	**RMSE**	**MAE**	**DA**
Technical Indicators	0,091	0,075	0,533
Technical Indicators with Lagged Returns	0,091	0,076	0,530
XGBoost	**RMSE**	**MAE**	**DA**
Technical Indicators	0,090	0,075	0,530
Technical Indicators with Lagged Returns	0,090	0,075	0,550
SVR	**RMSE**	**MAE**	**DA**
Technical Indicators	0,090	0,075	0,533
Technical Indicators with Lagged Returns	0,090	0,076	0,533
LSTM	**RMSE**	**MAE**	**DA**
Technical Indicators	0,149	0,123	0,500

Table A26: TRV Model Results

TRV			
Random Forest	**RMSE**	**MAE**	**DA**
Technical Indicators	0,064	0,054	0,400
Technical Indicators with Lagged Returns	0,063	0,053	0,577
XGBoost	**RMSE**	**MAE**	**DA**
Technical Indicators	0,065	0,056	0,400
Technical Indicators with Lagged Returns	0,063	0,053	0,555
SVR	**RMSE**	**MAE**	**DA**
Technical Indicators	0,066	0,056	0,377
Technical Indicators with Lagged Returns	0,061	0,050	0,533
LSTM	**RMSE**	**MAE**	**DA**
Technical Indicators	0,121	0,092	0,386

Table A27: UNH Model Results

UNH			
Random Forest	**RMSE**	**MAE**	**DA**
Technical Indicators	0,062	0,048	0,466
Technical Indicators with Lagged Returns	0,061	0,048	0,422
XGBoost	**RMSE**	**MAE**	**DA**
Technical Indicators	0,062	0,049	0,488
Technical Indicators with Lagged Returns	0,062	0,049	0,488
SVR	**RMSE**	**MAE**	**DA**
Technical Indicators	0,051	0,047	0,488
Technical Indicators with Lagged Returns	0,059	0,046	0,0466
LSTM	**RMSE**	**MAE**	**DA**
Technical Indicators	0,107	0,085	0,454

Table A28: V Model Results

V			
Random Forest	**RMSE**	**MAE**	**DA**
Technical Indicators	0,061	0,05	0,548
Technical Indicators with Lagged Returns	0,059	0,049	0,466
XGBoost	**RMSE**	**MAE**	**DA**
Technical Indicators	0,059	0,048	0,486
Technical Indicators with Lagged Returns	0,057	0,049	0,466
SVR	**RMSE**	**MAE**	**DA**
Technical Indicators	0,058	0,046	0,580
Technical Indicators with Lagged Returns	0,066	0,053	0,333
LSTM	**RMSE**	**MAE**	**DA**
Technical Indicators	0,058	0,045	0,500

Table A29: VZ Model Results

VZ			
Random Forest	**RMSE**	**MAE**	**DA**
Technical Indicators	0,059	0,048	0,480
Technical Indicators with Lagged Returns	0,059	0,047	0,577
XGBoost	**RMSE**	**MAE**	**DA**
Technical Indicators	0,056	0,045	0,533
Technical Indicators with Lagged Returns	0,058	0,047	0,377
SVR	**RMSE**	**MAE**	**DA**
Technical Indicators	0,060	0,049	0,422
Technical Indicators with Lagged Returns	0,062	0,049	0,466
LSTM	**RMSE**	**MAE**	**DA**
Technical Indicators	0,057	0,046	0,723

Table A30: WMT Model Results

WMT			
Random Forest	**RMSE**	**MAE**	**DA**
Technical Indicators	0,057	0,042	0,600
Technical Indicators with Lagged Returns	0,057	0,043	0,555
XGBoost	**RMSE**	**MAE**	**DA**
Technical Indicators	0,057	0,056	0,400
Technical Indicators with Lagged Returns	0,059	0,045	0,511
SVR	**RMSE**	**MAE**	**DA**
Technical Indicators	0,057	0,044	0,660
Technical Indicators with Lagged Returns	0,055	0,044	0,581
LSTM	**RMSE**	**MAE**	**DA**
Technical Indicators	0,095	0,072	0,511

B. Sector-Based Model Results

Table B1: Technology-Based Model Results

Technology			
Random Forest	**RMSE**	**MAE**	**DA**
Technical Indicators	0,092	0,075	0,540
Technical Indicators with Lagged Returns	0,092	0,075	0,545
XGBoost	**RMSE**	**MAE**	**DA**
Technical Indicators	0,092	0,075	0,555
Technical Indicators with Lagged Returns	0,092	0,073	0,535
SVR	**RMSE**	**MAE**	**DA**
Technical Indicators	0,097	0,079	0,498
Technical Indicators with Lagged Returns	0,094	0,076	0,546
LSTM	**RMSE**	**MAE**	**DA**
Technical Indicators	0,133	0,109	0,540

Table B2: Financials-Based Model Results

Financials			
Random Forest	**RMSE**	**MAE**	**DA**
Technical Indicators	0,074	0,061	0,515
Technical Indicators with Lagged Returns	0,074	0,062	0,427
XGBoost	**RMSE**	**MAE**	**DA**
Technical Indicators	0,088	0,073	0,482
Technical Indicators with Lagged Returns	0,076	0,064	0,444
SVR	**RMSE**	**MAE**	**DA**
Technical Indicators	0,074	0,061	0,523
Technical Indicators with Lagged Returns	0,074	0,061	0,345
LSTM	**RMSE**	**MAE**	**DA**
Technical Indicators	0,133	0,103	0,496

Table B3: Consumer Discretionary-Based Model Results

Consumer Discretionary			
Random Forest	**RMSE**	**MAE**	**DA**
Technical Indicators	0,084	0,177	0,524
Technical Indicators with Lagged Returns	0,084	0,176	0,537
XGBoost	**RMSE**	**MAE**	**DA**
Technical Indicators	0,084	0,065	0,428
Technical Indicators with Lagged Returns	0,176	0,067	0,528
SVR	**RMSE**	**MAE**	**DA**
Technical Indicators	0,090	0,071	0,533
Technical Indicators with Lagged Returns	0,081	0,198	0,515
LSTM	**RMSE**	**MAE**	**DA**
Technical Indicators	0,107	0,241	0,508

Table B4: Industrials-Based Model Results

Industrials			
Random Forest	**RMSE**	**MAE**	**DA**
Technical Indicators	0,091	0,073	0,477
Technical Indicators with Lagged Returns	0,092	0,073	0,478
XGBoost	**RMSE**	**MAE**	**DA**
Technical Indicators	0,090	0,071	0,483
Technical Indicators with Lagged Returns	0,088	0,070	0,488
SVR	**RMSE**	**MAE**	**DA**
Technical Indicators	0,124	0,097	0,533
Technical Indicators with Lagged Returns	0,103	0,080	0,516
LSTM	**RMSE**	**MAE**	**DA**
Technical Indicators	0,101	0,081	0,499

Table B5: Healthcare-Based Model Results

Healthcare			
Random Forest	**RMSE**	**MAE**	**DA**
Technical Indicators	0,061	0,048	0,516
Technical Indicators with Lagged Returns	0,061	0,048	0,516
XGBoost	**RMSE**	**MAE**	**DA**
Technical Indicators	0,062	0,049	0,533
Technical Indicators with Lagged Returns	0,061	0,048	0,516
SVR	**RMSE**	**MAE**	**DA**
Technical Indicators	0,061	0,049	0,499
Technical Indicators with Lagged Returns	0,065	0,051	0,428
LSTM	**RMSE**	**MAE**	**DA**
Technical Indicators	0,082	0,065	0,514

C. ARIMA

The Autoregressive Integrated Moving Average (ARIMA) model is a traditional statistical time series model that uses the time series' own historical values and error terms to make future predictions (Box & Jenkins, 1976). Because stock prices are not stationary, it is often applied to return series (Tsay, 2010). It has been used as a fundamental tool in financial forecasting for many years and has yielded reasonable results in short-term portfolios (Adebiyi et al., 2014).

ARIMA on US Market

The ARIMA model alone cannot demonstrate high performance when applied to US markets. For S&P 500 index, the model is often treated as a random walk (Uzzal and Ślepaczuk, 2023). Performance is even weaker during periods of high volatility and complexity and forecasts drift rapidly, particularly due to market shocks and structural breaks (Wang, 2024).

Limitations

ARIMA cannot understand nonlinear relationships due to its linearity and stationarity assumptions (Tsay, 2010). The model struggles to account for complex, and volatility clusters and requires alternative hybrid models. Due to its linear, univariate structure, it relies on the

series' own history, and its performance can deteriorate when market conditions change (Wang, 2024).

ARIMA vs. Machine Learning Approaches

Studies in the literature indicate that ML methods are superior to ARIMA (Wang, 2024). While ARIMA struggles to capture nonlinear relationships, as seen in this study, different ML models have demonstrated high predictive performance across different stocks and have been able to adapt to different market conditions.

While RF performed better than ARIMA in short-term forecasting, LSTM performed better in long-term forecasting with lower error than ARIMA. LSTM is capable of learning long-term dependencies better than ARIMA (Wang, 2024).

Advantages and Disadvantages

ARIMA has the advantage of being simpler and more straightforward than other time-series models, requiring less data (Box & Jenkins, 1976). Its disadvantages include its incompatibility with linear approaches, its inability to capture complexity and volatility, its univariate structure, and its inability to capture sectoral trends (Tsay, 2010).